**18. Three Entrances** A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

| Entrance | 1 | 2 | 3 |
|---|---|---|---|
| Number Entering | 83 | 61 | 56 |

Q18

$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$ $H_1$: at least 1 is diff

$n = 83 + 61 + 56 = 200$

$E = 200 \times \frac{1}{3} = 66.67$

$\chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67}$

$\chi^2 \approx 6.19$

$df = 2$

R.R. $\chi^2_{0.05;2} \approx 5.99$

Since $6.19 > 5.99$, we rej $H_0$.

$\hat{p} = \frac{83}{200} = 0.415$

C.I.: $\hat{p} \pm z \cdot SE = 0.415 \pm 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$= 0.415 \pm 1.96 \times \sqrt{\frac{0.415(0.585)}{200}}$

$= (0.3468, 0.4832)$

**22. Heart Attacks on Mondays** Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

| Day | Observed Count |
|---|---|
| Sunday | 24 |
| Monday | 36 |
| Tuesday | 27 |
| Wednesday | 26 |
| Thursday | 32 |
| Friday | 26 |
| Saturday | 29 |

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0$. $p_1 = p_2 = \dots = p_7 = \frac{1}{7}$ $H_1$: at least 1 is diff

$n = 200$

$E = \frac{200}{7} \approx 28.57$

$\chi^2 = \frac{(24-28.57)^2}{28.57} + \frac{(36-28.57)^2}{28.57} + \frac{(27-28.57)^2}{28.57} + \frac{(26-28.57)^2}{28.57} + \frac{(32-28.57)^2}{28.57} + \frac{(26-28.57)^2}{28.57} + \frac{(29-28.57)^2}{28.57}$

$\approx 3.629$

$\alpha = 0.05$ $df = 6$

R.R. $\chi^2_{0.05;6} = 12.592$

Since $3.629 < 12.592$, we fail to rej. $H_0$.

state, describe the nature of these differences.

**18. Fatal Accidents** Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

| | Small | Medium | Large | |
|---|---|---|---|---|
| Fatal | 67 | 26 | 16 | 109 |
| Not Fatal | 128 | 63 | 46 | 237 |
| | 195 | 89 | 62 | 346 |

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

$H_0$: classifications are independent $H_1$: classifications are dependent

$E_{1,1} = \frac{109 \times 195}{346} \approx 61.43$ $E_{2,1} = \frac{237 \times 195}{346} \approx 133.57$

$E_{1,2} = \frac{109 \times 89}{346} \approx 28.04$ $E_{2,2} = \frac{237 \times 89}{346} \approx 60.96$

$E_{1,3} = \frac{109 \times 62}{346} \approx 19.53$ $E_{2,3} = \frac{237 \times 62}{346} \approx 42.47$

$$\chi^2 = \frac{(67-61.43)^2}{61.43} + \frac{(26-28.04)^2}{28.04} + \frac{(16-19.53)^2}{19.53} + \frac{(128-133.57)^2}{133.57} + \frac{(63-60.96)^2}{60.96} + \frac{(46-42.47)^2}{42.47}$$

$\chi^2 \approx 1.884$

$\alpha = 0.05$ $df = 1 \times 2 = 2$

R.R.

$\chi^2_{2;0.05} = 5.991$

Since $1.884 < 5.991$, we fail to rej. $H_0$.

**23. Telecommuting** As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

DATA SET DS1403

| | Workdays at Home per Week | | | |
|---|---|---|---|---|
| Salary | Less Than One | At Least One, but Not All | All at Home | |
| Under $50,000 | 38 | 16 | 14 | 68 |
| $50,000–$74,999 | 54 | 26 | 12 | 92 |
| $75,000–$99,999 | 35 | 22 | 9 | 66 |
| Above $100,000 | 33 | 29 | 12 | 74 |
| | 160 | 93 | 47 | 300 |

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the $p$-value for this test of hypothesis. Does the $p$-value confirm your conclusions from part a?

a) $H_0$: salary independent of no. of workdays $H_1$: salary dependent

$E_{1,1} = \frac{(68 \times 160)}{300} \approx 36.27$ $E_{1,2} = \frac{(68 \times 93)}{300} \approx 21.08$ $E_{1,3} = \frac{(68 \times 47)}{300} \approx 10.653$

$E_{2,1} \approx 49.067$ $E_{2,2} \approx 28.52$ $E_{2,3} \approx 14.413$

$E_{3,1} \approx 35.2$ $E_{3,2} \approx 20.46$ $E_{3,3} \approx 10.34$

$E_{4,1} \approx 39.467$ $E_{4,2} \approx 22.94$ $E_{4,3} \approx 11.59$

$\chi^2 = 6.44$

$\alpha = 0.05$ $df = 3 \times 2 = 6$

R.R.

$\chi^2_{6;0.05} \approx 12.592$

Since $6.44 < 12.592$, we fail to rej $H_0$.

b) $df = 6$, $\chi^2 = 12.592$

$P(\chi^2 = 6.45) = 0.375 > 0.05$

$\therefore$ Fail to rej. $H_0$.